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Case Projects

Case Project 1-1: Attack Experiences

Incident Overview

In early 2025, a critical security incident occurred on a workstation used for developing mobile applications and database prototypes. The attack was identified as a **Ransomware infection**. It took place shortly after a user downloaded what appeared to be a necessary driver utility from a third-party website while troubleshooting hardware compatibility issues.

Nature of the Attack and Damage

The attack was a payload-based ransomware that immediately began encrypting high-value file types, including .java, .kt, .php, and .sql files. This effectively locked several weeks of progress on academic and personal coding projects. In addition to data encryption, the malware disabled the Windows Task Manager and modified the boot configuration to prevent entering Safe Mode, rendering the system almost entirely unresponsive to standard recovery attempts.

Analysis of Success

Based on technical evaluation and common security frameworks, the attack was successful due to several intersecting vulnerabilities:

Factor	Reasoning for Success
Administrative Privileges	The user was logged into an account with full administrative rights, allowing the malware to execute system-level registry changes without further authorization.
Phishing/Social Engineering	The malicious site successfully mimicked a legitimate hardware support portal, leading the user to bypass standard security warnings.
Lack of Real-time Scanning	The installed antivirus definitions were outdated by several days, which allowed the newer ransomware signature to go undetected during the initial download.

Recovery and Resolution

The computer was fixed by performing a low-level format of the primary storage drive to ensure all malicious sectors were cleared. This was followed by a clean installation of the operating system. Data recovery was partially successful through the use of an external backup drive that was fortunately disconnected during the time of the attack, though approximately 15% of the most recent work was lost.

Preventative Measures

To prevent a recurrence, several new protocols have been established. This includes the use of a standard (non-admin) user account for daily development tasks, the implementation of "3-2-1" backup strategy (including automated cloud syncing), and the activation of behavioral-based endpoint detection software that can identify encryption patterns even if a specific virus signature is unknown.

"This experience highlights the critical need for constant vigilance and the implementation of multiple layers of security, particularly when handling sensitive development environments."